

Parallel Robots

EPFL – 2026

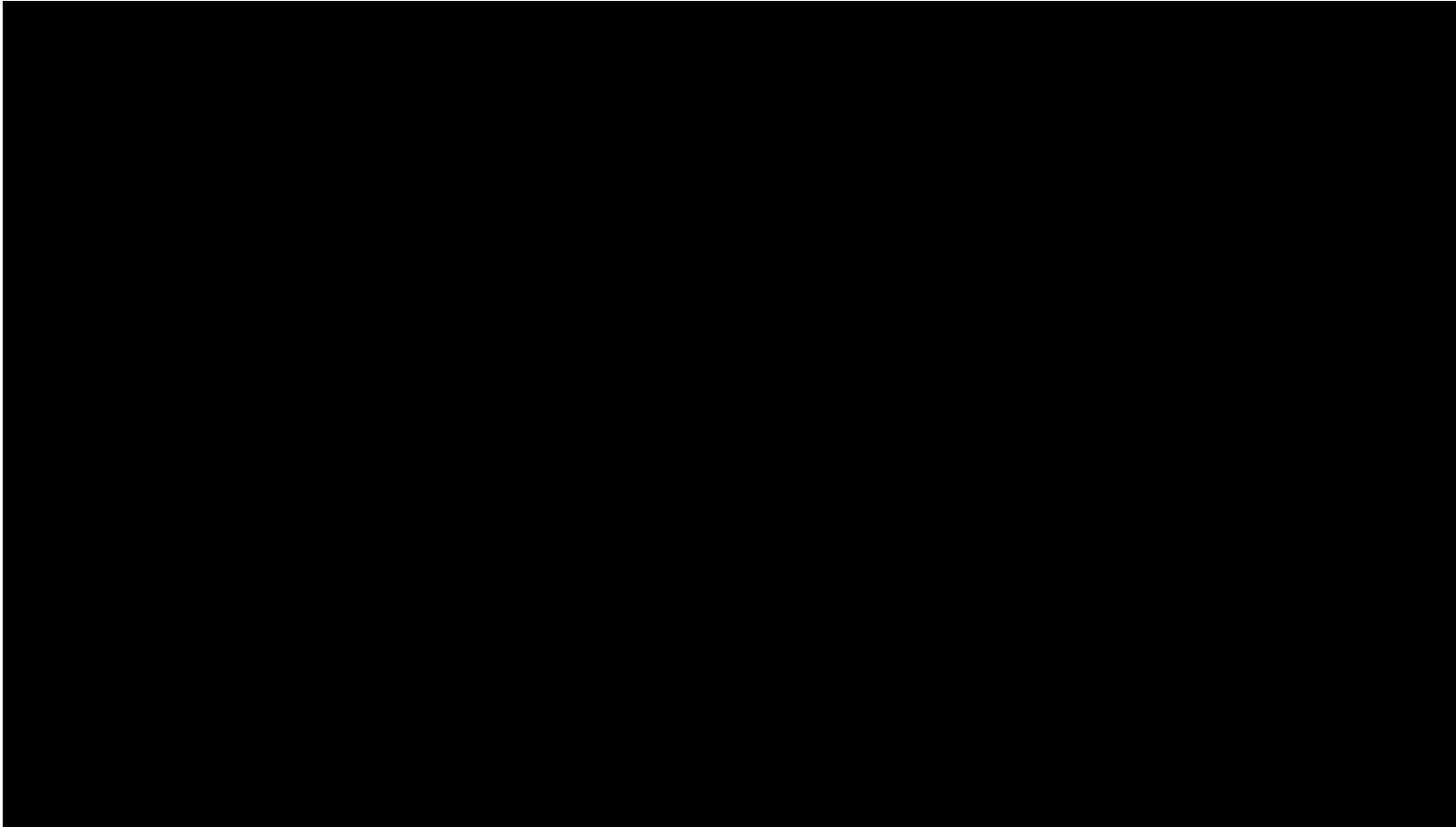
Arda YIĞIT



Redundancy in Robotics

- Important limitation of parallel robots: workspace
- Translational workspace
 - Possible to scale up
- Orientation workspace
 - Scale invariant
- What is limiting orientations?
- Mechanical interference
 - Between legs
- Joint range of motion
 - Ball and socket joint approx. cone angle 40°
- Singularities
 - Typically, the maximum tilt of a Gough-Stewart platform is approx. 45°

- How to enlarge orientation workspace?
- “Technological” solution
 - Clever mechanical design to avoid/reduce mechanical interference
 - Large range of motion joints

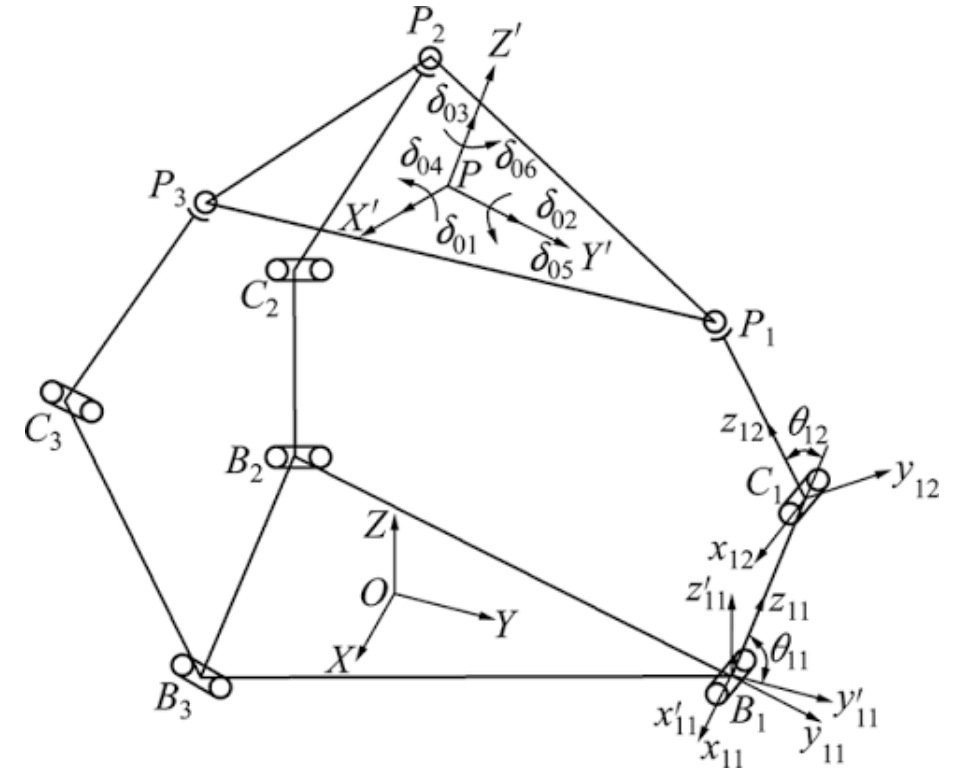


<https://youtu.be/f0hNbOkorRY>

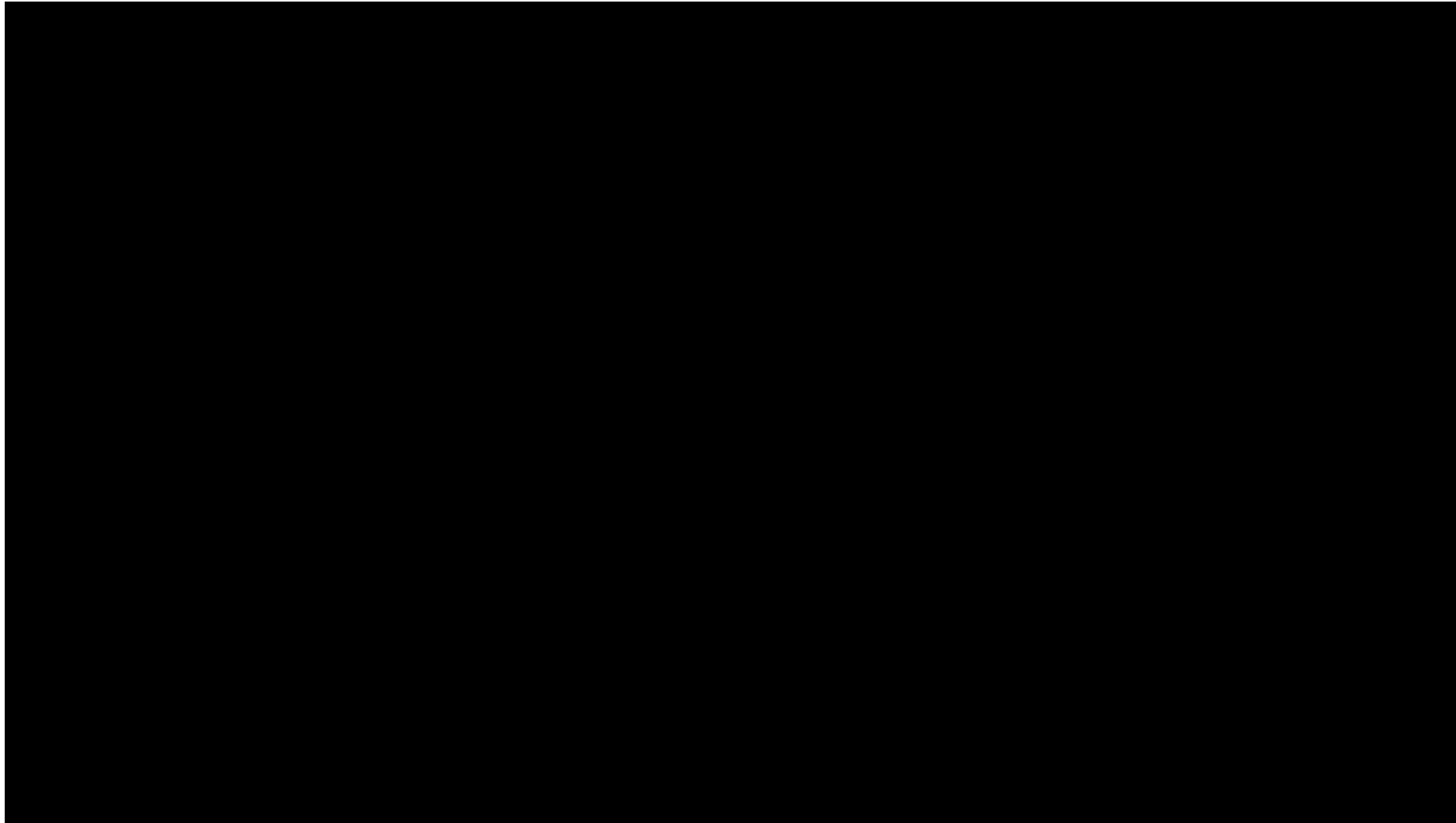
- How to enlarge orientation workspace?
- Introducing redundancy to avoid singularities
 - Kinematic redundancy
 - Actuation redundancy
- Let us see why and how redundancy allows for avoiding singularities

- Degree of freedom (DOF): the number of **independent coordinates** required to define the **task** specifically
- Often, it corresponds to a subgroup of end-effector pose
- We will see counter-examples later
- The DOF depends on the task
- The DOF is independent of the choice of coordinates

- Example 1: 5-bar mechanism
- Example 2: 3-RRS robot



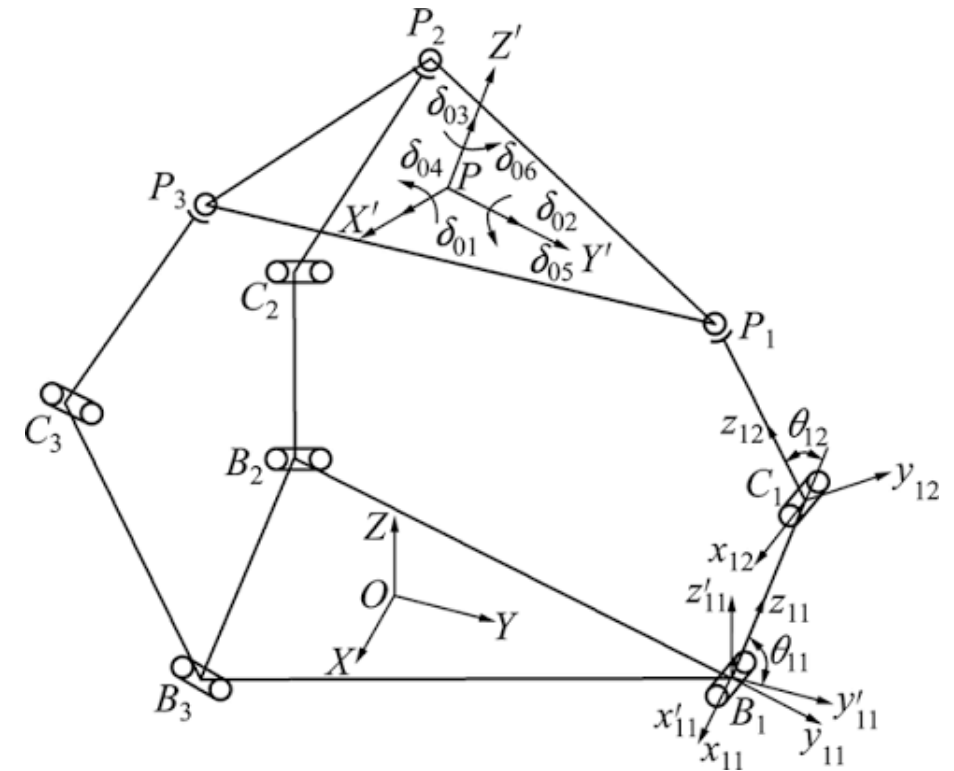
<https://doi.org/10.1109/MACE.2010.5535537>



https://youtu.be/Bf_1pUyjNiM

- Degree of mobility (DOM): the total number of **independent coordinates** required to define the overall **configuration** of the entire mechanism
- The DOM of a mechanism is unique
- Physical interpretation: the minimum number of elementary joints (R or P) that must be locked simultaneously to completely immobilize the entire mechanism

- Example 1: 5-bar mechanism
- Example 2: see previous video
- Example 3: 3-RRS robot



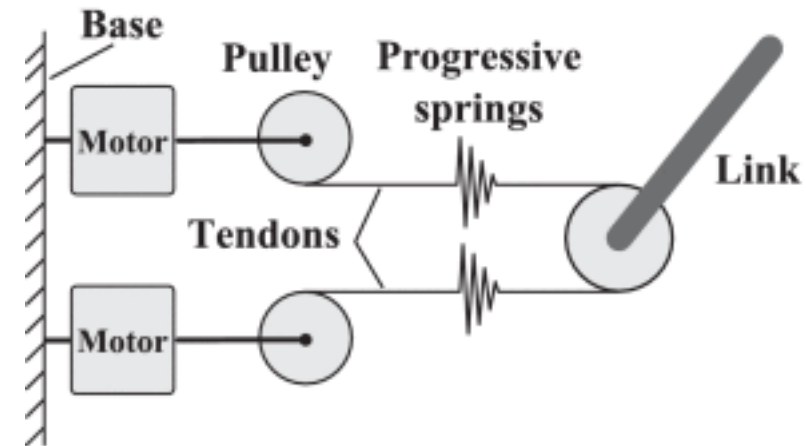
<https://doi.org/10.1109/MACE.2010.5535537>

$$M = d(n - g - 1) + \sum_{i=1}^g f_i$$

- M is the DOM
- $d = 3$ for planar mechanisms,
 $d = 6$ for spatial mechanisms
- n is the total number of links in the mechanism, including the base
- g is the total number of joints connecting the links
- f_i is the DOF permitted by joint i
- $M > 0$ means the mechanism can be controlled with M independent actuators
- $M = 0$ means the mechanism is static
- $M < 0$ means the mechanism is over-constrained
- **Warning:** this formula does not always give the correct solution, and it is not easy to anticipate when it will not work...

- Degree of actuation (DOA): the total number of **independent actuators** used to drive a parallel robot
- Equivalently, DOA is the total number of actively driven joints in the mechanism
- Is DOA meaningful for serial robots?
 - Variable stiffness actuators
 - Tendon-driven serial robots

- Under-actuated: $DOA < DOM$
 - Fully-actuated: $DOA = DOM$
 - Over-actuated: $DOA > DOM$
-
- Find examples for these 3 cases

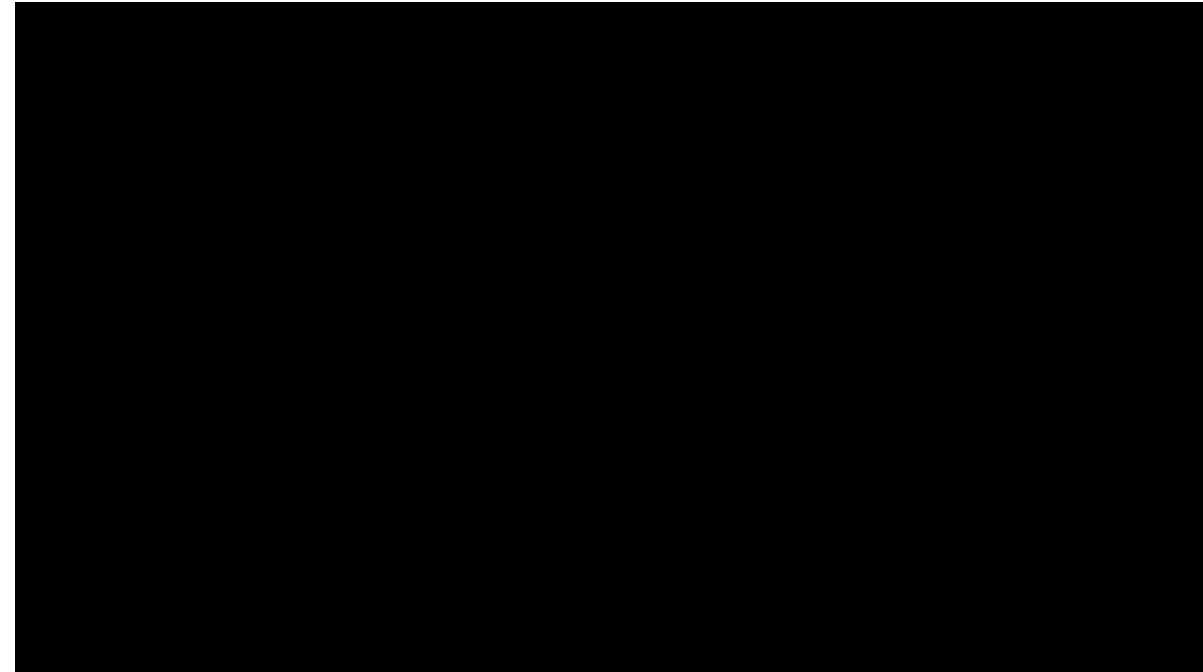


<https://doi.org/10.1109/TMECH.2015.2501019>

Kinematically-Redundant Parallel Robots

- A robot is **kinematically redundant** (KR) if $DOM > DOF$, i.e., its mobility exceeds what is strictly necessary for the task
- KR is not a static property of the robot itself, as it is defined relative to a task
- KR is often obtained by introducing **additional actuated joints** in one or more legs of the mechanism
- Example: modifying the 5-bar mechanism to make it kinematically redundant

- Active singularity avoidance
- Larger workspace
 - Consequence of singularity avoidance
- Remote actuation
- Optimizing secondary criteria
 - Stiffness maximization
 - Kinetic energy minimization
- Better force and torque distribution



<https://youtu.be/zxTSp4aRa-o>

- Let us consider $DOM = DOA$
- Joint coordinates $\mathbf{q} \in \mathbb{R}^{n_q}$
- Task coordinates $\mathbf{x} \in \mathbb{R}^{n_x}$
- $n_q = DOM, n_x = DOF$
- $\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$
- $\mathbf{A} \in \mathbb{R}^{n_x \times n_x}$
- $\mathbf{B} \in \mathbb{R}^{n_x \times n_q}$
- Why matrix $\mathbf{B}$ is not square?
- This kinematics mapping has an infinite number of solutions
- Two redundancy resolution techniques will be explored
 - Pseudoinverse
 - Augmented task space

- Let $\dot{\mathbf{x}}_d$ be the desired task velocity
- All solutions to

$$\mathbf{A}\dot{\mathbf{x}}_d = -\mathbf{B}\dot{\mathbf{q}}$$

are of the form

$$\dot{\mathbf{q}} = -\mathbf{B}^+ \mathbf{A}\dot{\mathbf{x}}_d + (\mathbf{I} - \mathbf{B}^+ \mathbf{B})\mathbf{z}$$

for any $\mathbf{z} \in \mathbb{R}^{n_q}$ and with $\mathbf{B}^+$ the Moore-Penrose pseudoinverse

$$\mathbf{B}^+ = \mathbf{B}^T (\mathbf{B}\mathbf{B}^T)^{-1}$$

- Consider the augmented task-space coordinates

$$\mathbf{x}_{aug} = \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_r \end{bmatrix} \in \mathbb{R}^{n_q}$$

- What is the dimension of $\mathbf{x}_r$?
- How to choose $\mathbf{x}_r$?
- Differential kinematics become square

- Let us consider $\mathbf{A}_r$ and $\mathbf{B}_r$ such that
$$\mathbf{A}_r \dot{\mathbf{x}}_r + \mathbf{B}_r \dot{\mathbf{q}} = \mathbf{0}$$

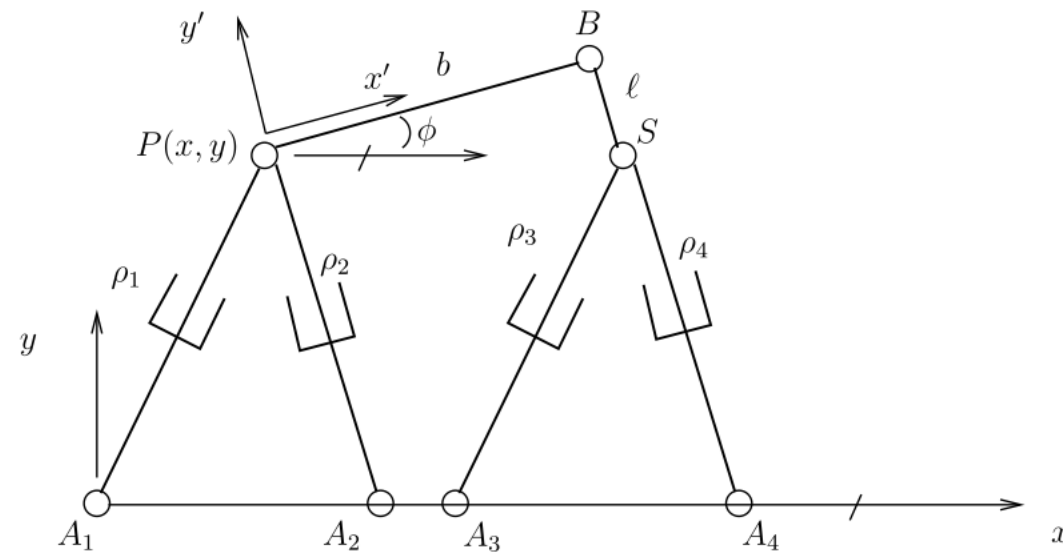
- What are the dimensions of $\mathbf{A}_r$ and $\mathbf{B}_r$?

- Differential kinematics with augmented task coordinates are

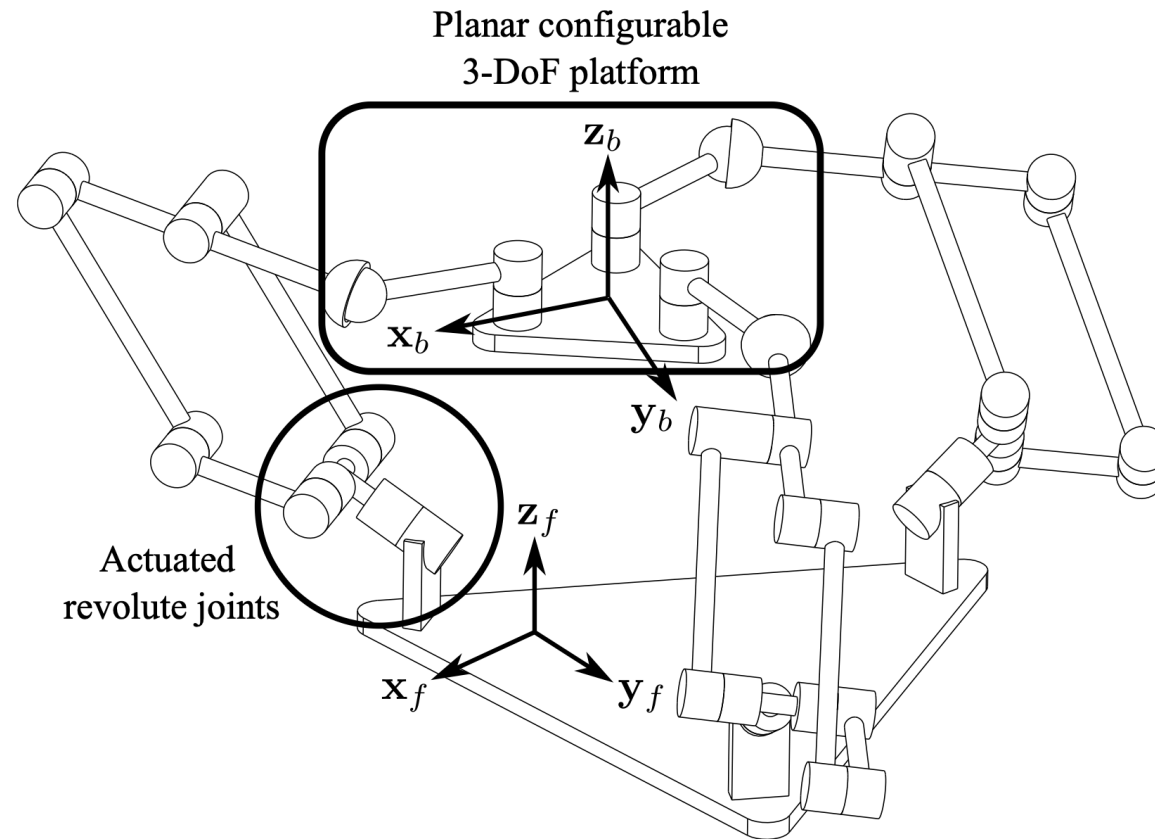
$$\mathbf{A} \dot{\mathbf{x}} + \mathbf{A}_r \dot{\mathbf{x}}_r + \begin{bmatrix} \mathbf{B} \\ \mathbf{B}_r \end{bmatrix} \dot{\mathbf{q}} = \mathbf{0}$$

- What are the expressions of $\mathbf{A}_{aug}$ and $\mathbf{B}_{aug}$?

- Planar mechanism from <https://doi.org/10.1109/TRO.2015.2409433>



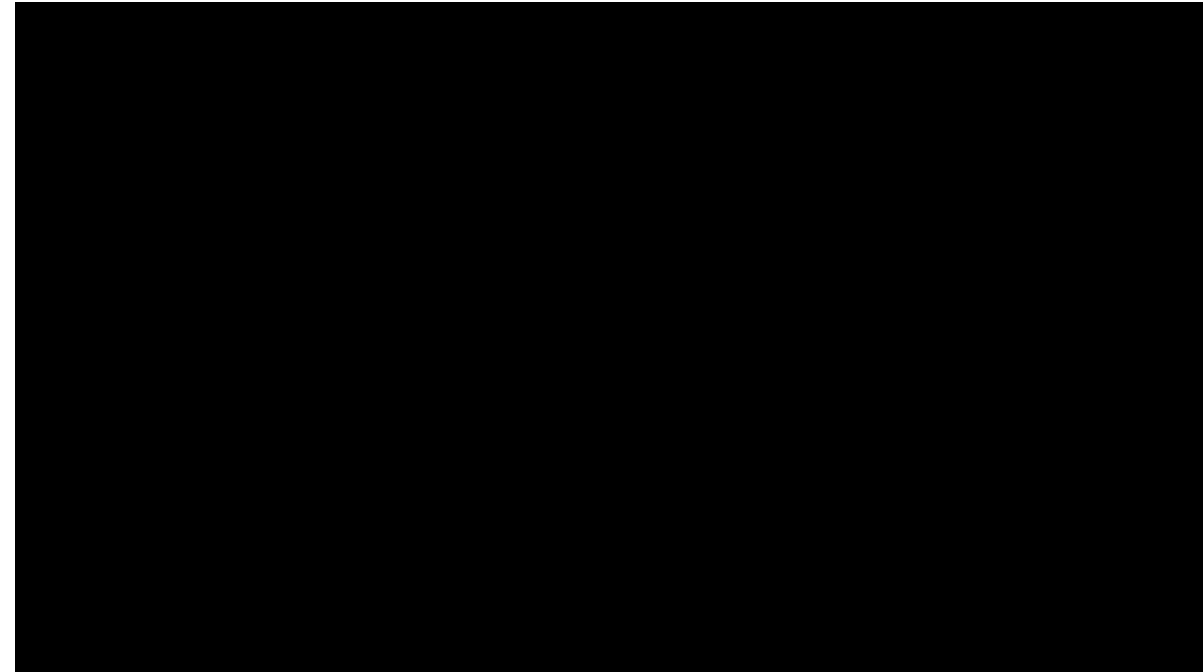
- Spatial mechanism from <https://doi.org/10.1115/1.4063407>



Actuation-Redundant Parallel Robots

- A robot is **actuation-redundant** (AR) if $DOA > DOM$, i.e., the number of actuated joints exceeds the mobility
- Unlike KR, AR is an intrinsic structural/actuation property of the robot, independent of the task coordinates
- AR is often obtained by **adding actuators** to existing legs or **adding extra legs** to a nonredundant parallel mechanism
- Example: modifying the 5-bar mechanism to make it actuation redundant

- Elimination of singularities
- Larger workspace
 - Consequence of singularity removal
- Internal force control and backlash clearance
- Enhanced payload and force distribution



https://youtu.be/tFHIV_bfrDI

- Let us consider $DOF = DOM$
- Joint coordinates $\mathbf{q} \in \mathbb{R}^{n_q}$
- Task coordinates $\mathbf{x} \in \mathbb{R}^{n_x}$
- $n_q = DOA, n_x = DOM$
- $\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$
- $\mathbf{A} \in \mathbb{R}^{n_q \times n_x}$
- $\mathbf{B} \in \mathbb{R}^{n_q \times n_q}$
- Why matrix $\mathbf{A}$ is not square?
- This kinematics mapping has eventually no solution
- A statics-based redundancy resolution technique will be explored

- Statics formulation for desired end-effector wrench $\mathbf{w}_d$

$$\mathbf{J}_{inv}^T \boldsymbol{\tau} = \mathbf{w}_d$$

with $\mathbf{J}_{inv}^T = -(\mathbf{B}^{-1}\mathbf{A})^T$. $\mathbf{J}_{inv}^T$ is a wide matrix (i.e., more columns than rows)

- General solution is

$$\boldsymbol{\tau} = (\mathbf{J}_{inv}^T)^+ \mathbf{w}_d + \left(\mathbf{I} - (\mathbf{J}_{inv}^T)^+ \mathbf{J}_{inv}^T \right) \boldsymbol{\lambda}$$

for any $\boldsymbol{\lambda} \in \mathbb{R}^{n_q}$ and with $(\mathbf{J}_{inv}^T)^+$ the Moore-Penrose pseudoinverse

$$(\mathbf{J}_{inv}^T)^+ = \mathbf{J}_{inv} (\mathbf{J}_{inv}^T \mathbf{J}_{inv})^{-1}$$

- Planar 4-RPR mechanism

- Spatial 3-PPPS mechanism